

Fraction Detective File-Folder Mystery

Unit 2: Fraction Detective Agency • Lesson 2-1 • Flagship • EduWonderLab

UNIT 2 • LESSON 2-1 ★ FLAGSHIP • TEACHER NOTES

Standard 6.NS.1

FORMAT Model-to-equation mystery	TIME 30–35 min
GROUPING Teams of 3–4	TOPIC Interpret Division of Fractions (Flagship)
STANDARD 6.NS.1	PREP LEVEL Medium — print model clues into file folders

Learning goal *I can match a model to the fraction division equation it represents and create my own.*

Teacher setup

- Print one mystery packet per team. Each clue describes a model.
- Teams decide which division equation each model shows.
- The final clue asks the team to create their own matching model.

Materials

File-folder model clues, Fraction strips or bar models, Blank model card, Pencil.

Differentiation & language support

- Provide pre-cut fraction strips so teams can rebuild each model (SPED).
- Sentence stem: “This model shows $\frac{\quad}{\quad} \div \frac{\quad}{\quad}$ because $\frac{\quad}{\quad}$.”
- ESOL: label each model with “how many fit?” or “share equally?”

Key vocabulary (English / Español):

Term	Español	What it means
model	modelo	a picture that shows the math
equation	ecuación	a number sentence with an equal sign
represent	representar	to stand for

Quick teacher check: *Check that the final model for $3 \div 1/2$ shows 6 equal half-pieces.*

Fraction Detective File-Folder Mystery

Unit 2: Fraction Detective Agency • Lesson 2-1 • Flagship • EduWonderLab

Name: _____

Date: _____

Class: _____

Your goal: *I can match a model to the fraction division equation it represents and create my own.*

What you need

Model clue cards, Fraction strips, Blank model card.

How to play

1. Open a model clue and rebuild it with fraction strips if needed.
2. Decide which division equation the model represents.
3. Find the quotient and write what it means.
4. Solve the final clue by creating your own model.

Model Clues

Read each model clue, decide which division equation it shows, and find the quotient.

- | |
|--|
| 1. Clue A: Two whole bars are each cut into thirds. Count the thirds. |
| 2. Clue B: A $\frac{3}{4}$ bar is split into 3 equal parts. |
| 3. Clue C: Four whole bars, marked into $\frac{2}{3}$ -size groups. |
| 4. Clue D: One half bar is cut into eighths. Count the eighths. |
| 5. Clue E: Six cups filled by $\frac{3}{4}$ -cup scoops. |
| 6. Final Clue: Draw your own model for $3 \div \frac{1}{2}$ and write what the quotient means. |

Detective log (equation + meaning for each clue):

Explain your thinking

Explain how a model can show the difference between sharing and measuring out groups.

Sentence frame: *This model shows ___ $\div$ ___ because it asks how many ___.*

★ Challenge: *Create a second model that looks different but shows the same equation as Clue E.*

ANSWER KEY

Teacher copy — please do not distribute to students.

UNIT 2 • LESSON 2-1 ★ FLAGSHIP • Fraction Detective File-Folder
Mystery

Standard 6.NS.1

Answers

1. Clue A: Two whole bars are each cut into thirds. Count the thirds.

Answer: $2 \div \frac{1}{3} = 6$

2. Clue B: A $\frac{3}{4}$ bar is split into 3 equal parts.

Answer: $\frac{3}{4} \div 3 = \frac{1}{4}$

3. Clue C: Four whole bars, marked into $\frac{2}{3}$ -size groups.

Answer: $4 \div \frac{2}{3} = 6$

4. Clue D: One half bar is cut into eighths. Count the eighths.

Answer: $\frac{1}{2} \div \frac{1}{8} = 4$

5. Clue E: Six cups filled by $\frac{3}{4}$ -cup scoops.

Answer: $6 \div \frac{3}{4} = 8$

6. Final Clue: Draw your own model for $3 \div \frac{1}{2}$ and write what the quotient means.

Answer: 6 — there are 6 half-pieces in 3 wholes

Teaching notes & look-fors

- Clue E quotient is 8 (six cups $\div$ $\frac{3}{4}$ -cup scoops).
- Look for models where students confuse $6 \div \frac{3}{4}$ with $\frac{3}{4} \div 6$.